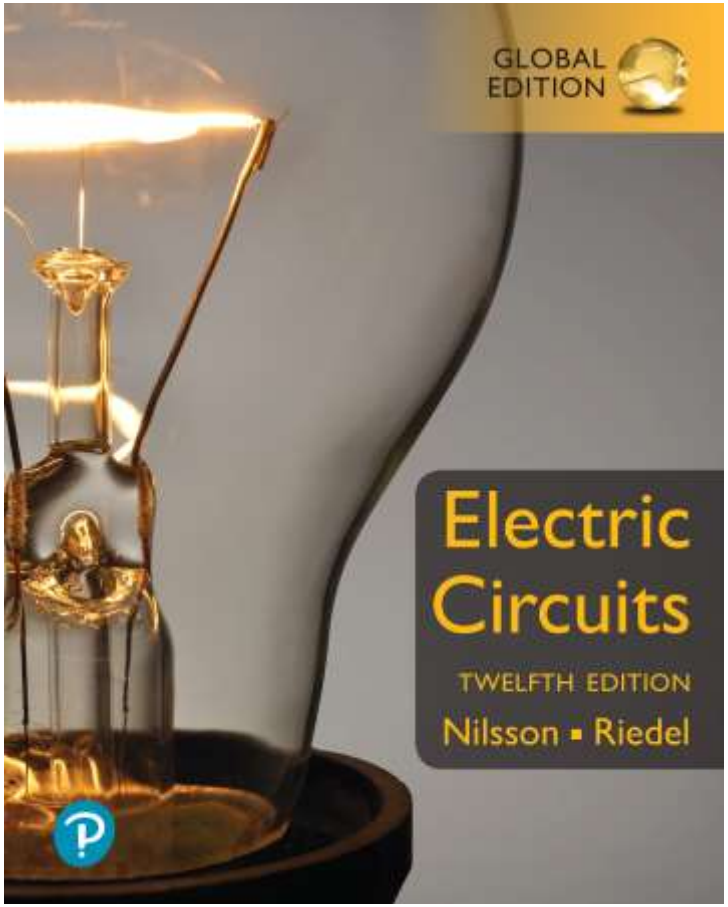
Electric Circuits

Twelfth Edition, Global Edition



Chapter 9 – Part 3

Sinusoidal Steady-State Analysis

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Chapter 9 – Sinusoidal Steady-State Analysis (1 of 26)

Steps in ACSS Analysis:

1. Redraw the circuit (the phasor transform does not change the components or their connections).
2. Phasor transform all known v(t) and i(t).
3. Represent unknown voltages and currents with *V* and *I*.
4. Replace component values with impedance (Z) values.
5. Use any circuit analysis method(s) to write equations and solve them with a calculator.
6. Inverse-transform the result, which is a phasor, back to the time domain.

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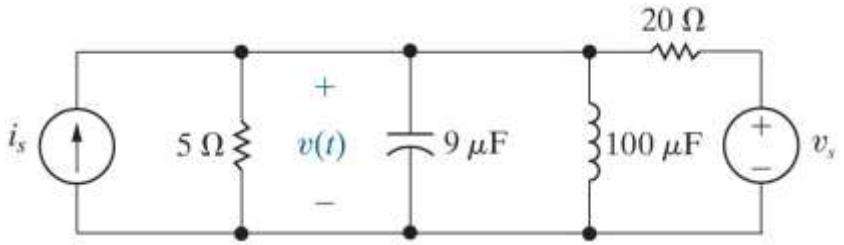
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Chapter 9 – Sinusoidal Steady-State Analysis (2 of 26)

Example:

$i_s(t) = 10\cos(50{,}000t)$ A;

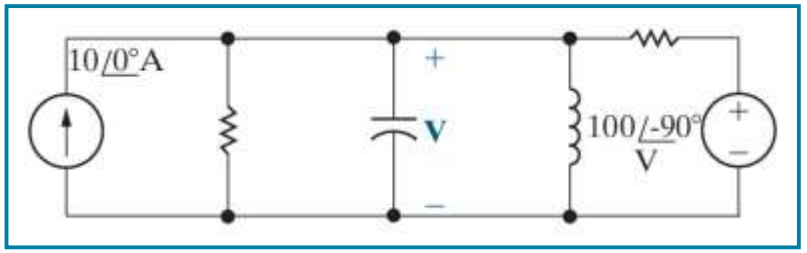
$v_s(t) = 100\sin(50{,}000t)$ V.



Find $v(t)$ in the steady-state.

1. Redraw the circuit.
2. Phasor transform all known voltages and currents.
3. Represent unknown voltages and currents with phasor symbols

$$v_s(t) = 100\sin 50{,}000t \text{ V}$$
$$= 100\cos(50{,}000t - 90°) \text{ V}$$



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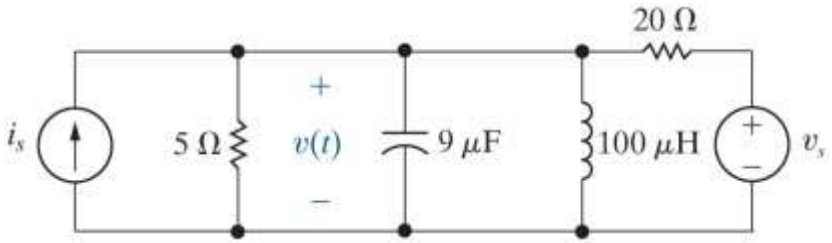
Chapter 9 – Sinusoidal Steady-State Analysis (3 of 26)

Example (continued):

$i_s(t) = 10\cos(50{,}000t)$ A;

$v_s(t) = 100\sin(50{,}000t)$ V.

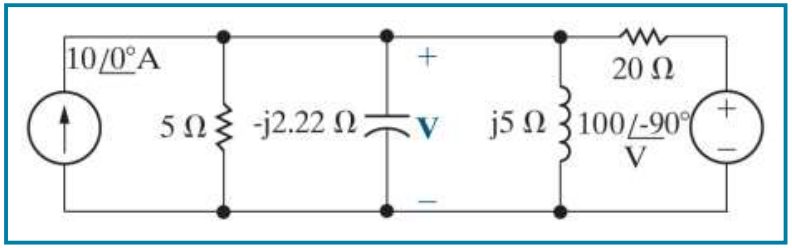
Find $v(t)$ in the steady-state.



4. Calculate the impedances of all resistors, inductors, and capacitors.

$$Z_L = j\omega L = j(50{,}000)(100\mu) = j5\,\Omega$$

$$Z_C = \frac{-j}{\omega C} = \frac{-j}{(50{,}000)(9\mu)} = -j2.22\,\Omega$$

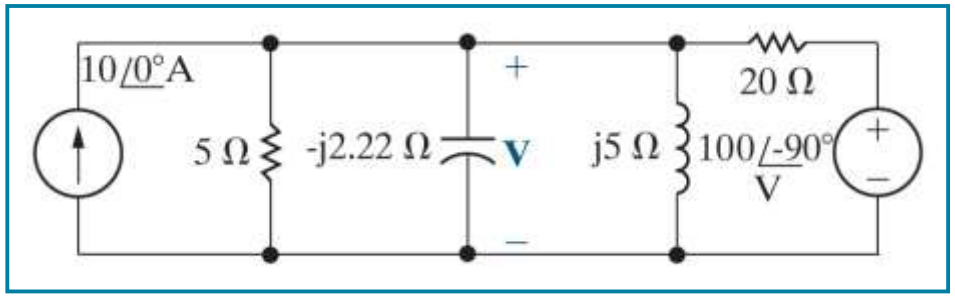


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Chapter 9 – Sinusoidal Steady-State Analysis (4 of 26)

Which of the following circuit analysis techniques provides a one-equation method to find the phasor voltage?



✗ A. Mesh current method

✗ B. Source transformation

✓ C. Node voltage method

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Chapter 9 – Sinusoidal Steady-State Analysis (5 of 26)

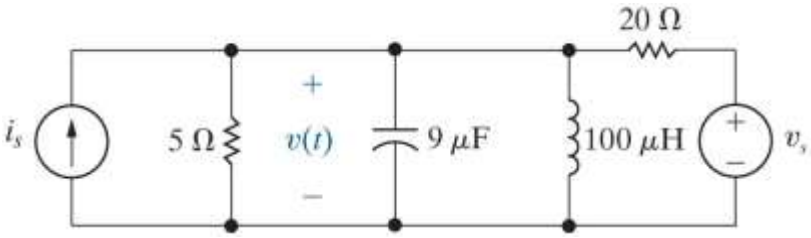
Example (continued):

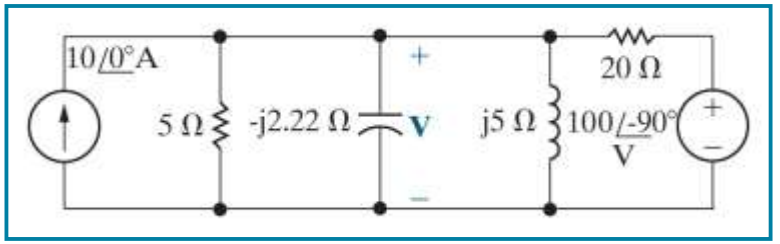
$i_s(t) = 10\cos(50{,}000t)$ A;

$v_s(t) = 100\sin(50{,}000t)$ V.

Find $v(t)$ in the steady-state.

5. Use DC circuit analysis techniques to find the phasor $\mathbf{V}$.
   - Use the node voltage method.





$$-10\angle 0° + \frac{\mathbf{V}}{5} + \frac{\mathbf{V}}{-j2.22} + \frac{\mathbf{V}}{j5} + \frac{\mathbf{V} - j100}{20} = 0$$

$$\mathbf{V}\left(\frac{1}{5} + \frac{1}{-j2.22} + \frac{1}{j5} + \frac{1}{20}\right) = 10 + j5$$

$$\therefore \quad \mathbf{V} = 10 - j30 \ V$$

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Chapter 9 – Sinusoidal Steady-State Analysis (6 of 26)

In writing the equations previously, we replaced the complex number $100\angle -90°$ with –j100. Your calculator can do this for you by putting its representation of complex numbers in

- ✔ A. Rectangular form
- ✘ B. Polar form
- ✘ C. Approximate form

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Chapter 9 – Sinusoidal Steady-State Analysis (7 of 26)

Convert the complex number -15 to polar form.

- ✗ A. $-15\angle 0°$
- ✗ B. $15\angle -90°$
- ✓ C. $15\angle 180°$

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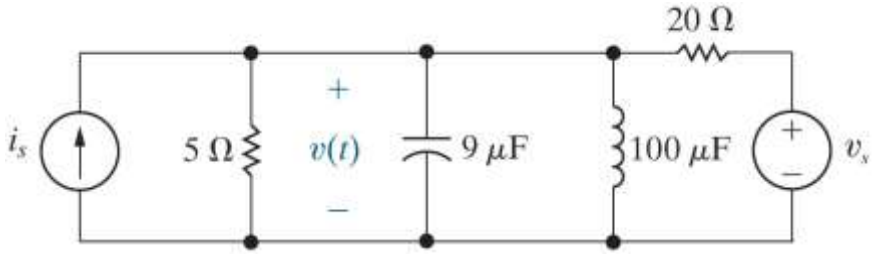
Chapter 9 – Sinusoidal Steady-State Analysis (8 of 26)

Example (continued):

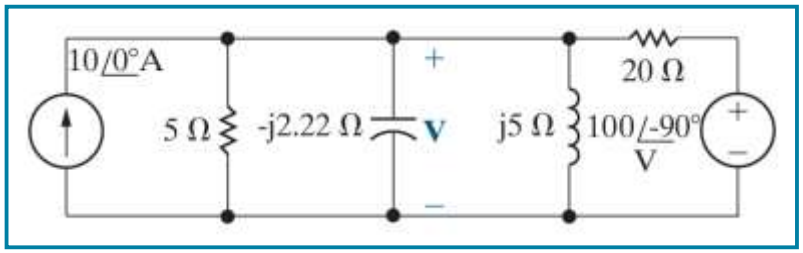
$i_s(t) = 10\cos(50{,}000t)$ A;

$v_s(t) = 100\sin(50{,}000t)$ V.

Find $v(t)$ in the steady-state.



6. Inverse phasor-transform the result to get back to the time *domain*.



$$\mathbf{V} = 10 - j\,30\,V = 31.62\angle -71.57°V$$

$$\mathcal{P}^{-1}\{31.62\angle -71.57°\} = 31.62\cos(50{,}000t - 71.57°)\ V$$

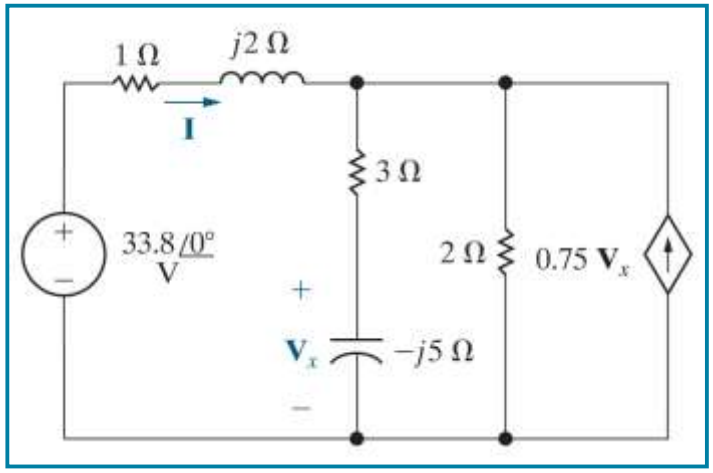
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Chapter 9 – Sinusoidal Steady-State Analysis (9 of 26)

Example:

Find the value of the phasor current *I* in the phasor-transformed circuit shown.



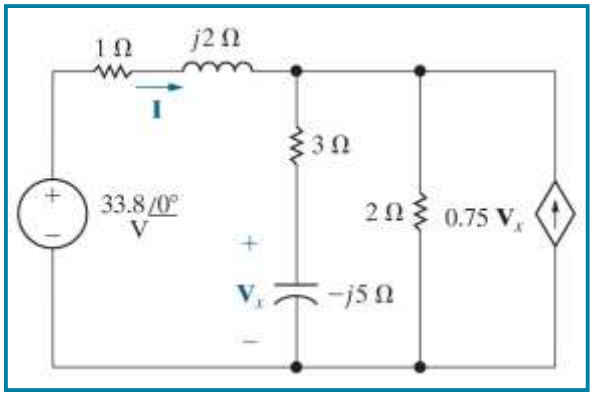
5. Use DC circuit analysis techniques to find the phasor ***I***.

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Chapter 9 – Sinusoidal Steady-State Analysis (10 of 26)

Which of the circuit analysis methods below will generate equations that, when solved, directly result in the value of the phasor current?



✗ A. Node voltage method

✓ B. Mesh current method

✗ C. Source transformation

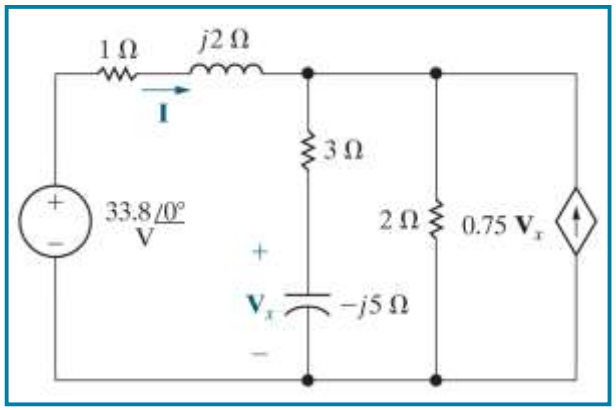
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Chapter 9 – Sinusoidal Steady-State Analysis (11 of 26)

Example (continued):

Find the value of the phasor current *I* in the phasor-transformed circuit shown.



Mesh current equations:

$$-33.8 + (1 + j2)\mathbf{I} + (3 - j5)(\mathbf{I} - \mathbf{I}_1) = 0$$

$$(3 - j5)(\mathbf{I}_1 - \mathbf{I}) + 2(\mathbf{I}_1 + 0.75\mathbf{V}_x) = 0$$

$$\mathbf{V}_x = -j5(\mathbf{I} - \mathbf{I}_1)$$

Standard form:

$$\mathbf{I}(1 + j2 + 3 - j5) + \mathbf{I}_1(-3 + j5) + \mathbf{V}_x(0) = 33.8$$

$$\mathbf{I}(-3 + j5) + \mathbf{I}_1(3 - j5 + 2) + \mathbf{V}_x[2(0.75)] = 0$$

$$\mathbf{I}(-j5) + \mathbf{I}_1(j5) + \mathbf{V}_x(-1) = 0$$

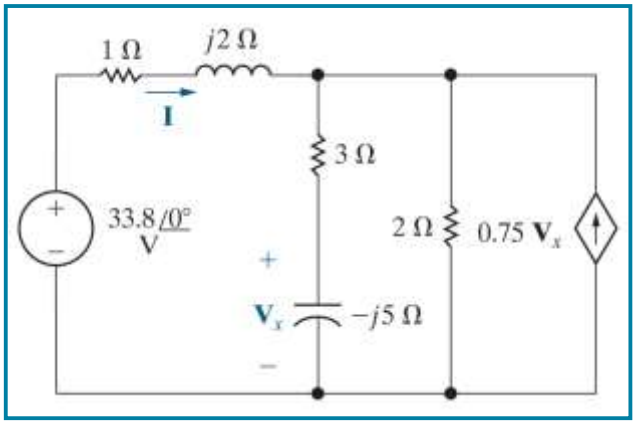
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Chapter 9 – Sinusoidal Steady-State Analysis (12 of 26)

Example (continued):

Find the value of the phasor current *I* in the phasor-transformed circuit shown.



Solving:

$$\mathbf{I} = 29 + j2\ A$$

$$\mathbf{I}_1 = 19.4 + j6\ A$$

$$\mathbf{V}_x = -20 + j48\ V$$

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Chapter 9 – Sinusoidal Steady-State Analysis (13 of 26)

In preparation for the inverse phasor transform, we need to convert the phasor current to

- ✗ A. Rectangular form
- ✓ B. Polar form
- ✗ C. Standard form

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Chapter 9 – Sinusoidal Steady-State Analysis (14 of 26)

If we convert 29 + j2 to polar form, its phase angle will be

- ✔ A. Positive
- ✘ B. Negative
- ✘ C. I need my calculator!

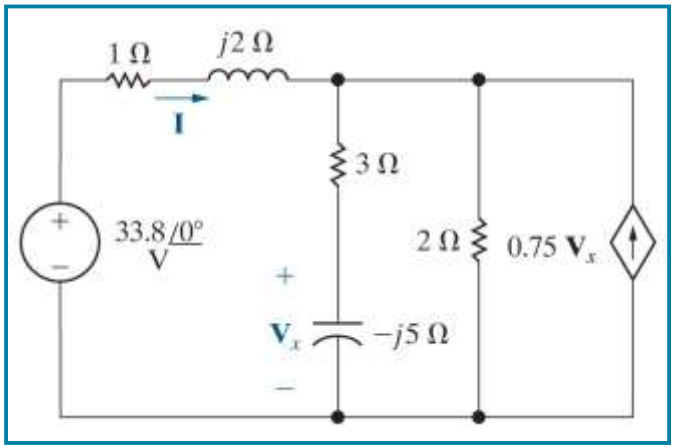
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Chapter 9 – Sinusoidal Steady-State Analysis (15 of 26)

Example (continued):

Find the value of the phasor current *I* in the phasor-transformed circuit shown.



6. Inverse phasor-transform the result to get back to the time domain. We don't have to complete this step for this problem!

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Chapter 9 – Sinusoidal Steady-State Analysis (16 of 26)

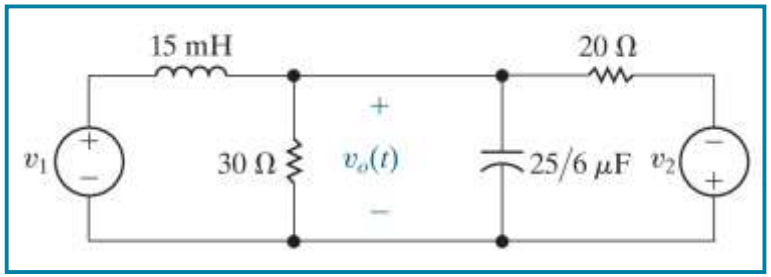
Example:

$v_1(t) = 240\cos(4000t + 53.13°)$ V;

$v_2(t) = 96\sin(4000t)$ V. Find

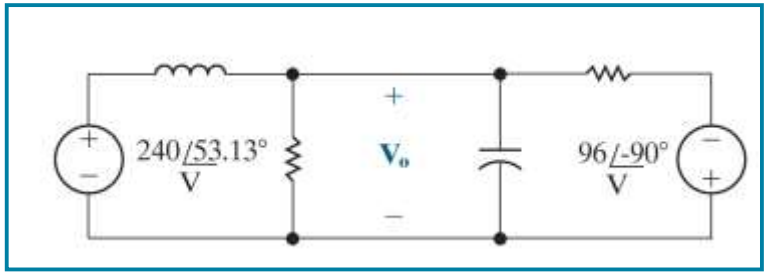
$v_o(t)$ in the steady-state, using a series of source transformations.

1. Redraw the circuit.
2. Phasor transform all known voltages and currents.
3. Represent unknown voltages and currents with phasor symbols



$$v_2(t) = 96\sin 4000t \ V$$

$$= 96\cos(4000t - 90°) \ V$$



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Chapter 9 – Sinusoidal Steady-State Analysis (17 of 26)

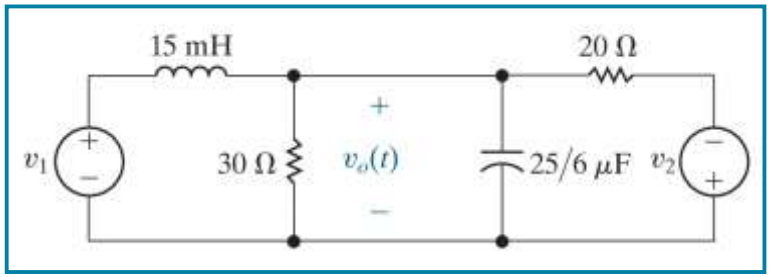
Example (continued):

$v_1(t) = 240\cos(4000t + 53.13°)$ V;

$v_2(t) = 96\sin(4000t)$ V. Find

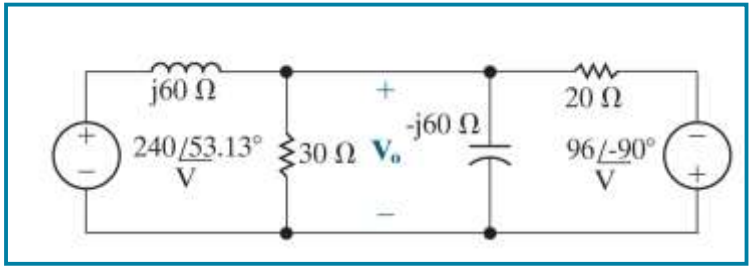
$v_o(t)$ in the steady-state, using a series of source transformations.

4. Calculate the impedances of all resistors, inductors, and capacitors.



$$Z_L = j\omega L = j(4000)(0.015) = j60\Omega$$

$$Z_C = -j/\omega C = -j/(4000)(^{25}/_{6}\,\mu) = -j60\Omega$$



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Chapter 9 – Sinusoidal Steady-State Analysis (18 of 26)

We can check our impedance values to be sure that the resistor impedances are _____ and the capacitor impedances are _____.

- ✗ A. Positive real numbers; negative real numbers
- ✓ B. Positive real numbers; negative imaginary numbers
- ✗ C. Positive imaginary numbers; negative imaginary numbers

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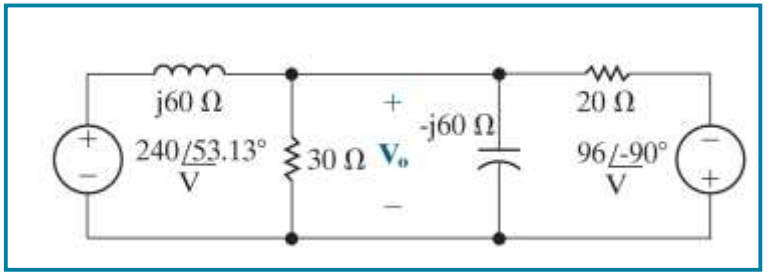
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Chapter 9 – Sinusoidal Steady-State Analysis (19 of 26)

Example (continued):

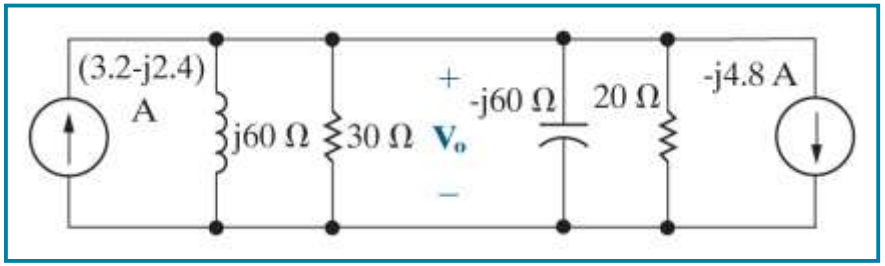
$v_1(t) = 240\cos(4000t + 53.13°)$ V; $v_2(t) = 96\sin(4000t)$ V. Find $v_o(t)$ in the steady-state, using a series of source transformations.

5. Source transform



$$\frac{240\angle 53.13°}{j60} = 3.2 - j2.4\ A$$

$$\frac{96\angle -90°}{-j60} = -j4.8\ A$$



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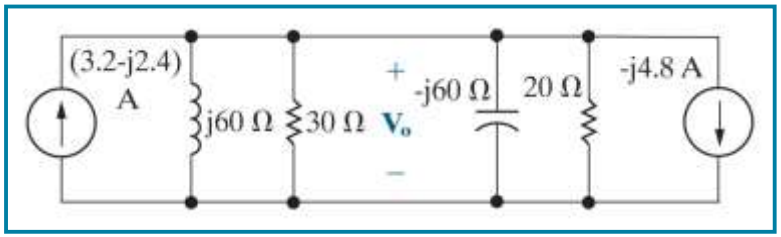
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Chapter 9 – Sinusoidal Steady-State Analysis (20 of 26)

Example (continued):

$v_1(t) = 240\cos(4000t + 53.13°)$ V; $v_2(t) = 96\sin(4000t)$ V. Find $v_o(t)$ in the steady-state, using a series of source transformations.

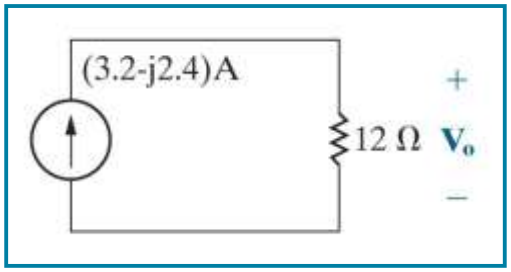
5. Source transform



$$3.2 - j2.4 - (-j4.8) = 3.2 - j2.4\ A$$

$$j60 \,||\, 30 \,||\, -j60 \,||\, 20 = 12\Omega$$

$$\mathbf{V}_o = (3.2 + j2.4)(12) = 38.4 + j28.8\ V$$



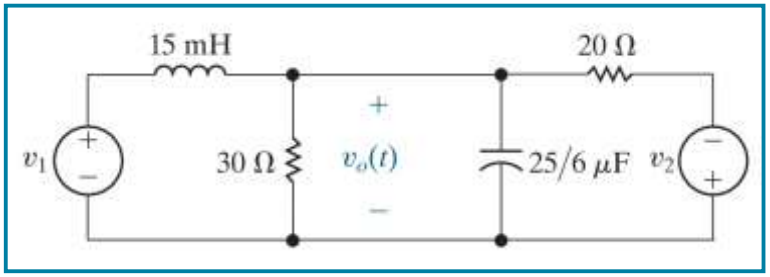
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Chapter 9 – Sinusoidal Steady-State Analysis (21 of 26)

Example (continued):

$v_1(t) = 240\cos(4000t + 53.13°)$ V; $v_2(t) = 96\sin(4000t)$ V. Find $v_o(t)$ in the steady-state, using a series of source transformations.



6. Inverse phasor-transform the result to get back to the time *domain*.

$$\mathbf{V}_o = 38.4 + j28.8 = 48\angle 36.87° \ V$$

$$v_o(t) = \mathscr{P}^{-1}\{\mathbf{V}_o\} = 48\cos(4000t + 36.87°) \ V$$

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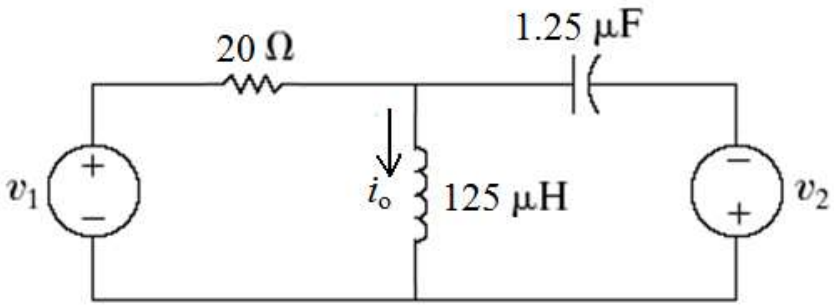
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Chapter 9 – Sinusoidal Steady-State Analysis (22 of 26)

Find the expression for $v_o(t)$ in the circuit below, given

$$v_1(t) = 5 \cos 80{,}000t \text{ V}, \quad \text{and}$$

$$v_2(t) = -2.5 \cos 320{,}000t \text{ V}$$

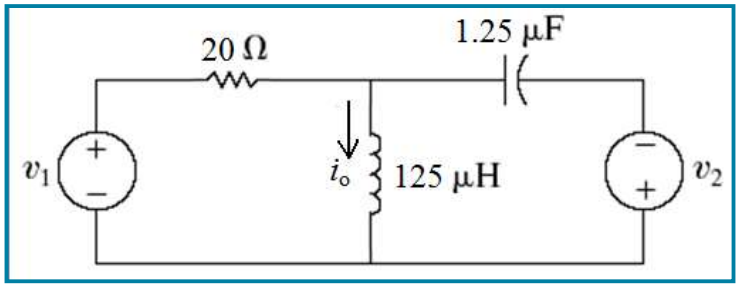


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Chapter 9 – Sinusoidal Steady-State Analysis (23 of 26)

Which formula should we use to calculate the impedance of the inductor? Remember that $v_1(t) = 5 \cos 80{,}000t$ V, and $v_2(t) = -2.5 \cos 320{,}000t$ V



- ✗ A. $j(80{,}000)(125\times10^{-6})$
- ✗ B. $j(320{,}000)(125\times10^{-6})$
- ✗ C. The average of A and B
- ✓ D. Both A and B

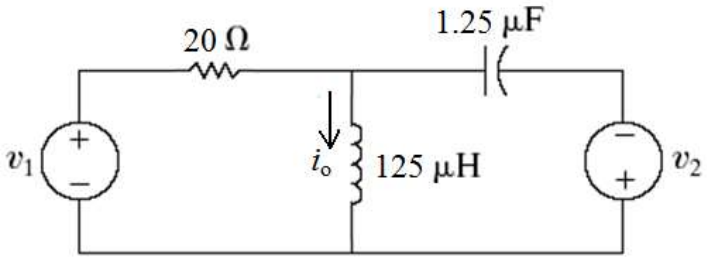
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Chapter 9 – Sinusoidal Steady-State Analysis (24 of 26)

Find the expression for $v_o(t)$ in the circuit below, given

$v_1(t) = 5 \cos 80{,}000t$ V, and $v_2(t) = -2.5 \cos 320{,}000t$ V



When the time domain sinusoidal sources have different frequencies, we must use superposition, phasor transforming and analyzing the circuit for each source individually.

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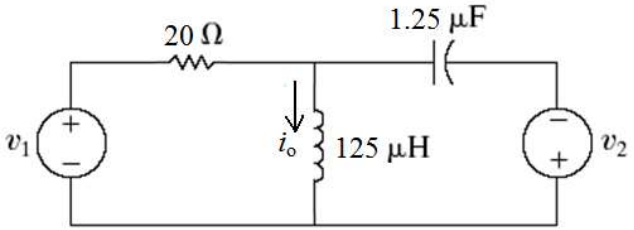
Chapter 9 – Sinusoidal Steady-State Analysis (25 of 26)

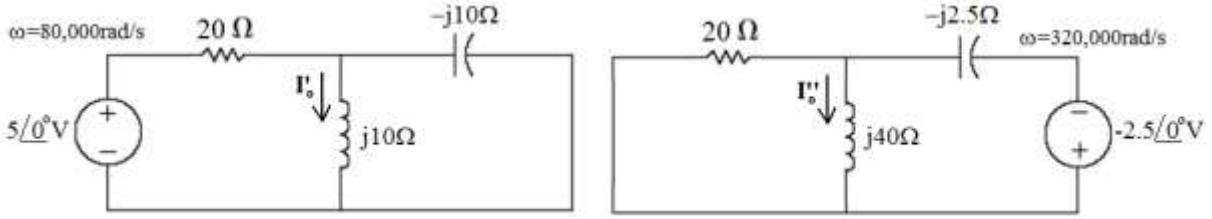
Find the expression for $v_o(t)$

in the circuit below, given

$v_1(t) = 5\cos 80{,}000t\ \text{V}$, and

$v_2(t) = -2.5\cos 320{,}000t\ \text{V}$





After source transform and current division,

$$\mathbf{I}_o' = -j500\text{mA} = 500\angle -90°\text{mA} \qquad \mathbf{I}_o'' = (8.73 - j65.5)\text{mA} = 66.08\angle -82.4°\text{mA}$$

Inverse phasor transform each current phasor, then add the two time domain expressions:

$$i(t) = 500\cos(80{,}000t - 90°) + 66.08\cos(320{,}000t - 82.4°)\text{mA}$$

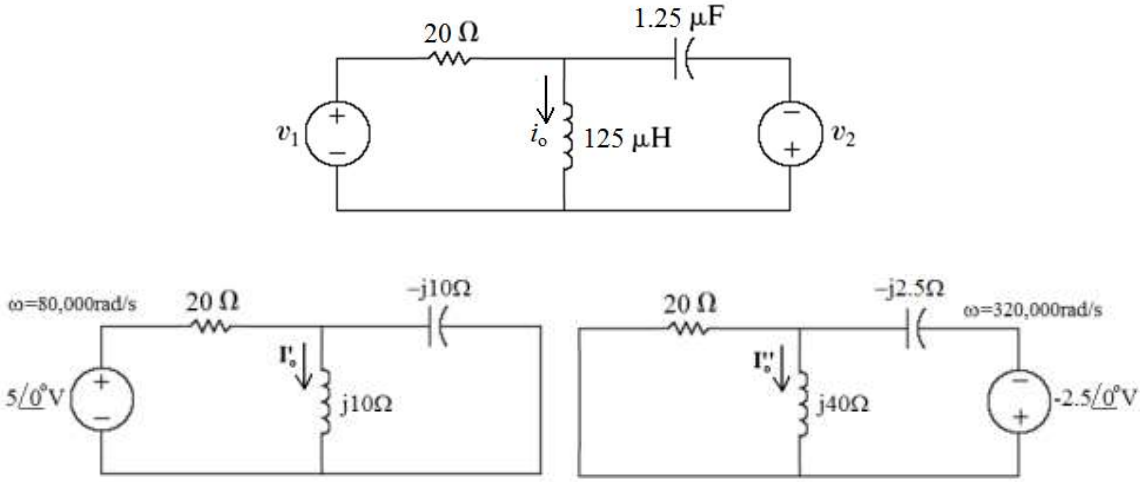
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Chapter 9 – Sinusoidal Steady-State Analysis (26 of 26)

Find the expression for $v_o(t)$ in the circuit below, given

$v_1(t) = 5\cos 80{,}000t$ V, and $v_2(t) = -2.5\cos 320{,}000t$ V



After source transform and current division,

$$\mathbf{I}_o' = -j500\text{mA} = 500\angle -90°\text{mA} \quad \mathbf{I}_o'' = (8.73 - j65.5)\text{mA} = 66.08\angle -82.4°\text{mA}$$

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